PAPER_964: 3D MUGE Magnetar Simulation

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Abstract

We construct a 3D MUGE magnetar simulation with three layers: (1) SCm superconducting core with radial BCS gap $\Delta(r) = \Delta_0(1 - (r/R)^2)$; (2) Abrikosov-type magnetic vortex tubes at $B > B_{\text{crit}} = 4.4 \times 10^{13}$ T; (3) 26-state phonon resonance shells at $R_n = R_{\text{NS}}(1 + 0.05n)$.

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1. SCm Core

$$\Delta(r) = \Delta_0 \left(1 - \frac{r^2}{R_{\text{core}}^2}\right), \quad r < R_{\text{core}}$$

2. Magnetic Vortex Tubes

$$n_v = \frac{B}{\Phi_0}, \quad \Phi_0 = \frac{h}{2e} \approx 2.068 \times 10^{-15} \text{ Wb}$$

3. Phonon Resonance Shells

$$R_n = R_{\text{NS}}(1 + 0.05n), \quad E_n = E_0(2\pi)^{n/3} S_{26}$$

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)